LESSON PLAN

PYRO TECHNIQUES (34)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Guided system, requirements and types based on stationary / moving targets
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Guided system, requirements and types based on stationary / moving targets
5. Specific Objectives	34.1.1.1 Trainee will be able to explain various terms on guided weapon.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of guided weapon terminology using IB and PDFs; Instructor supervision required for safety during discussion of live fire scenarios.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a guided weapon?
(b) Define the term 'guidance' in the context of ordnance.
(c) List three common types of guided missiles.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of guided weapon systems • Key terminology: Seeker, Bus, Warhead • Guidance laws: Proportional Navigation • Sensor types: Radar, IR, Laser • Signal processing basics	L & Ds	PPT, IB	15 min
(b)	• Requirements for stationary target engagement • Requirements for moving target engagement • Impact of target velocity on guidance • Lock-on modes: Pre-launch vs In-flight • Missile kinematics and terminal phase	L & Ds	PPT, PDF	20 min
(c)	• Classification by target type: Stationary • Classification by target type: Moving • Specific examples: Surface-to-Air vs Surface-to-Surface • Challenges in cluttered environments • Future trends in target acquisition	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference in guidance requirements between stationary and moving targets.
(b) Describe how a seeker head identifies a moving target amidst background clutter.
(c) Differentiate between pre-launch and in-flight lock-on modes with examples.

12. Home Assignment and Follow Up	Prepare a brief note comparing the guidance requirements for engaging a stationary tank versus a moving aircraft, referencing concepts from LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Control system, type and requirements	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min